

JOYANTA DEBNATH

AMIEB

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EDUCATION

Indian Institute of Technology Bombay <i>M.Tech in Remote Sensing, Department of Civil Engineering</i>	<i>Mumbai, India</i> 2026 – Present (1st Semester)
Indian Institute of Engineering Science and Technology, Shibpur <i>Bachelor of Technology in Civil Engineering</i> CGPA: 7.54 (First Class)	<i>West Bengal, India</i> 2021 – 2025
Govt. Science College, Tejgaon <i>Higher Secondary School Certificate (HSC)</i> GPA: 5.00 / 5.00	<i>Dhaka, Bangladesh</i> 2020
Sher-e-Bangla Nagar Govt. Boys' High School <i>Secondary School Certificate (SSC)</i> GPA: 5.00 / 5.00	<i>Dhaka, Bangladesh</i> 2018

WORK EXPERIENCE

Institute of Water Modelling (IWM) <i>Junior Engineer – Water Supply, Sanitation & Urban Management (WSU) Division</i>	<i>Dhaka, Bangladesh</i> July 2025 – July 2026
Urban Climate Resilient Infrastructure Project (UCRIP), Phase-2 (Client: LGED & KfW)	
Project Leader: Ahmad Salman Haider, Associate Specialist, WSU Division, IWM	
- Travelled to municipalities across Bangladesh to meet with local government officials and address issues related to urban infrastructure and resilience planning. - Collected field data on road conditions, community lifestyle, agricultural practices, and land-use issues to support project assessment and planning. - Coordinated with municipal officials to understand and document local infrastructure and resilience challenges for incorporation into project recommendations.	
Updating Sewerage Master Plan for Dhaka City (Client: UNICEF)	
Project Leader: Md. Sabbir Rahman, Associate Specialist, WSU Division, IWM	
- Executed end-to-end project activities ranging from comprehensive field surveys and data processing to final report submission. - Developed thematic maps and geospatial visualizations to support technical analysis of urban sanitation and sewerage planning. - Coordinated nighttime fecal sludge management (FSM) operations, overseeing sludge collection from residential areas and ensuring proper transport and testing at the designated Sewage Treatment Plant (STP).	
Extended Dhaka Water Supply Resilience Project (Client: DWASA)	
Project Leader: Tanmay Chaki, Director, WSU Division, IWM	
- Designed and implemented a custom Database Management System (DBMS) using Python to automate extensive data workflows. - Developed custom Python scripts to clean, standardize, and match complex household data across multiple District Metered Areas (DMAs). - Conducted Household Social and Existing Connection Surveys to gather and analyze critical demographic information, building typologies, and current water connection status for resilience planning.	
Detailed Design, Drawings & BOQ Preparation for Trust Green City Area (Client: Bangladesh Army)	
Project Leader: Tanmay Chaki, Director, WSU Division, IWM	
- Produced detailed 2D engineering drawings for drainage networks, pipe manholes, and utility systems using AutoCAD. - Created comprehensive 3D city models and architectural visualizations using SketchUp and Lumion to support integrated utility planning and presentation.	
Feasibility Study: Legacy Pollutant Removal from Buriganga River (Client: RID Unit, IWM)	
Supervisor: Md. Humayun Karim, Survey Specialist, Survey and Data Division, IWM	
- Conducted field surveys to collect water samples from the Buriganga River for environmental quality assessment. - Performed riverbed boring from bed level to sand level; collected clay and sand samples and prepared Bore Log Sheets.	

- Conducted geospatial map building, discharge data collection, and detailed statistical analysis of pollutant data to inform sustainable removal and disposal strategies.
- Conducted field sampling of local fish and cultivated vegetables along the Buriganga River to measure accumulated pollutants, coordinating with BCSIR for detailed laboratory analysis.

Dhaka City Water Supply Improvement Project (Client: Dhaka WASA)

Advisor: Amit Hassan Shishir, Sewer Network Engineer, WSU Division, IWM

- Developed detailed 2D engineering profiles and drawings using AutoCAD to support technical design and project analysis.

RESEARCH EXPERIENCE

Institute of Water Modelling (IWM)

Dhaka, Bangladesh

Research Intern | Supervisor: Tanmoy Chaki

May – June 2023

- Conducted dry-season water quality assessment of the Little Feni River using physicochemical and biological parameters; evaluated compliance with ECR 2023, IS 10500:2012, WHO, and EPA standards.
- Publication: Assessment of Water Quality in the Dry Season: A Case Study in the Little Feni River, Bangladesh. IOSR Journal of Environmental Science, Toxicology and Food Technology, June 2025. DOI: 10.9790/2402-1906016670

Indian Institute of Engineering Science and Technology (IIST), Shibpur

West Bengal, India

B.Tech Final Year Project | Supervisor: Prof. Debabrata Mazumder

Aug 2024 – May 2025

- Designed and evaluated a short-time aerobic composting system for organic waste management using vegetable peels; assessed decomposition efficiency and sustainability outcomes.
- DOI: 10.13140/RG.2.2.27427.16160

Institute of Water Modelling (IWM) – VARUNA Project

Dhaka, Bangladesh

Water Quality Index Calculator

June – July 2023

- Developed a Python-based system for automated computation and visualization of WQI, CWQI, and NSF indices for water resource assessment.
- Publication: Varuna: A Python-Based System for Water Quality Index Calculation. International Journal of Innovative Science and Research Technology, May 2025. DOI: 10.38124/ijisrt/25may2351

ACADEMIC PROJECTS & INTERNSHIPS

Institute for Steel Development and Growth (INSDAG)

India

Innovative Steel Structural Design – INSDAG 2023 (Awarded Best Design)

Supervised by Associate Professor Dr. Sujit Kumar Dalui

- Designed an earthquake-resistant and sustainable structure in a team setting.
- Developed conceptual, architectural, and structural plans; recognized as one of the Best Innovative Structural Designs of 2023.

Advanced Level Telecommunication Training Centre, BSNL

Ghaziabad, India

Project Intern (6 Weeks)

June – July 2024

- Conducted structural analysis of two buildings and foundation design using STAAD Pro.
- Performed RCC calculations for beams, columns, and slabs; designed architectural layouts and plumbing sections.

Structural Analysis Projects – IIST Shibpur (Architecture Collaboration)

May 2025

- AIFF National Centre of Excellence, New Town, Kolkata — collaborated with Indravo Ghosh.
- Multipurpose Indoor Sports Complex — collaborated with Pranav Vignesh.
- Conducted detailed structural analysis of RCC and steel structures for architectural design submissions using STAAD Pro.

TECHNICAL SKILLS

- Structural Analysis & Design: STAAD Pro
- Water Quality & Environmental Analysis: GIS, WHO/EPA Standards, Flowmaster Curve, SewerGems, SW DTM
- Architectural & 3D Modelling: AutoCAD, InfraWorks, Lumion, SketchUp
- Programming & Data Analysis: Python (Pandas, NumPy, Matplotlib), Excel, Microsoft Power BI

FELLOWSHIPS

ICCR Scholarship, India

- Awarded by the Indian Council for Cultural Relations (ICCR).
- Pursued B.Tech in Civil Engineering at IIST Shibpur under this scholarship; granted from the academic session 2021–22, covering the full course duration.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- General Secretary (Acting): Indian Knowledge Systems (IKS), IEST Shibpur
- Class Representative: Department of Civil Engineering, IEST Shibpur
- Main Co-Ordinator: Gangotri 1.0 (Indian Knowledge Systems, IEST Shibpur)
- Competitions & Seminars Head: REBECA 84th Edition (Annual Cultural Fest of IEST and oldest fest in India)
- Member: Anti-Ragging Committee, IEST Shibpur (Aug 2024 – May 2025)

LANGUAGES

English (Fluent) | Bengali (Native) | Hindi (Proficient)

REFERENCES

Prof. RAAJ Ramsankaran

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Prof. Eswar Rajasekaran

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